

第 1 章

現状

$$\left\{ \begin{array}{l} -\Delta u = \frac{1}{\epsilon^2}(-\alpha u + \beta u^2 - \gamma u^3 - \epsilon(av + bw)) \\ -\Delta v = u - v \\ -\Delta w = \frac{1}{d^2}(u - w) \\ \|u\|_{L^2} = N \end{array} \right. \quad (1.1)$$

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$$\mathbf{F}(\mathbf{u}, \epsilon) = Ku_h - \frac{-\alpha u_h + \beta u_h^2 - \gamma u_h^3 - \epsilon(aL(K + L)^{-1}L + bL(d^2K + L)^{-1})u_h}{\epsilon^2} \quad (1.3)$$

$$\mathcal{F}(\mathbf{X}) = \begin{pmatrix} \mathbf{F}(\mathbf{u}, \epsilon) \\ \sqrt{\mathbf{u}^T L \mathbf{u}} - N \end{pmatrix} = \mathbf{0} \quad (1.4)$$

$$\begin{pmatrix} \frac{\partial \mathbf{F}}{\partial \mathbf{u}} & \frac{\partial \mathbf{F}}{\partial \epsilon} \\ \frac{\partial g}{\partial \mathbf{u}} & \frac{\partial g}{\partial \epsilon} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{u} \\ \Delta \epsilon \end{pmatrix} = \begin{pmatrix} -\mathbf{F}(\mathbf{u}, \epsilon) \\ -g(\mathbf{u}) \end{pmatrix} \quad (1.5)$$

$$\frac{\partial \mathbf{F}}{\partial \mathbf{u}} = K - \frac{-\alpha L + 2\beta L u_h - 3\gamma L u_h^2 - \epsilon(aL(K + L)^{-1}L + bL(d^2K + L)^{-1}L)}{\epsilon^2} \quad (1.6)$$

$$\frac{\partial \mathbf{F}}{\partial \epsilon} = \frac{2(-\alpha L u_h + \beta u_h^2 - \gamma u_h^3)}{\epsilon^3} - \frac{(aL(K + L)^{-1}L + bL(d^2K + L)^{-1}L)u_h}{\epsilon^2} \quad (1.7)$$

$$\frac{\partial g}{\partial \mathbf{u}} = \frac{u^T L}{2\sqrt{u^T L u}} \quad (1.8)$$

$$\frac{\partial g}{\partial \epsilon} = 0 \quad (1.9)$$